

Metadoxy 50% w/w

For Animal Use Only

Soluble Powder

CONTENT: Each gm contains:
Doxycycline.....500.0 mg
(As doxycycline hyclate 577.1 mg)

PRODUCT DESCRIPTION: A yellowish crystalline powder.

PHARMACODYNAMICS:

Doxycycline binds reversibly to receptors of ribosomal fraction 30S, which leads to a blockade of the aminoacyl-tRNA binding to the corresponding site of the complex ribosome-RNA messenger. This results in inhibition of protein synthesis and therefore stops the growth of the bacterial culture. Doxycycline has a predominantly bacteriostatic activity. The bacteriostatic activity of doxycycline involves penetration of the substance into the bacterial cell. The penetration of doxycycline takes place by both active and passive diffusion. The main resistance mode is related to the possible presence of an R factor responsible for a decrease in the active transport of doxycycline. Doxycycline is a broad spectrum antibiotic. It is primarily active against Gram positive and negative microorganisms, aerobic and anaerobic, and against *Mycoplasma*, *Chlamydia* and *Rickettsiae*. Cross-resistance to other tetracyclines may occur. Usually such resistance has a plasmid origin. A continuous treatment with low doses of doxycycline may also result in an increased resistance to other antibiotics.

PHARMACOKINETICS:

Doxycycline is quickly (2-3 hours) absorbed after oral administration and its bioavailability in most species is around 70%. Doxycycline is highly bound to plasma proteins (about 90%). Highly fat soluble in comparison to first generation tetracyclines, doxycycline is widely distributed throughout the body. The highest concentrations were found in the lungs, kidneys, liver and spleen. Doxycycline crosses the placental barrier. Doxycycline is excreted by the biliary route, but a large proportion is reabsorbed by the small intestine (enterohepatic cycle). 40% of doxycycline is metabolized and excreted in the feces, mainly as inactive conjugated metabolites.

INDICATION:

In calves:

- Treatment and metaphylaxis of respiratory and digestive infections caused by micro-organisms susceptible to doxycycline.

In pigs:

- Treatment and metaphylaxis of respiratory infections caused by micro-organisms susceptible to doxycycline.

In chickens and turkeys:

- Treatment and metaphylaxis of respiratory infections due to micro-organisms susceptible to doxycycline.

The presence of the disease in the herd/flock should be established before metaphylactic treatment.

DOSAGE AND ADMINISTRATION:

To be administered orally in drinking water.

Calves and pigs:

10 mg of doxycycline per kg of body weight per day (equivalent to 11.54 mg of doxycycline hyclate/kg bw /day), during 3 to 5 days, or 0.2 g of powder per 10 kg of body weight per day, during 3-5 consecutive days, to be dissolved in drinking water.

Chickens and turkeys:

10 mg of doxycycline per kg of body weight per day (equivalent to 11.54 mg of doxycycline hyclate/kg bw /day), equivalent to 0.02 g of soluble powder per kg of body weight during 3 to 5 consecutive days, to be dissolved in drinking water.

The exact daily amount of oral powder based on the recommended dose and the number and weight of the animals to be treated should be calculated according to the following formula:

$$\frac{0.02 \text{ g of powder per kg of body weight per day}}{\text{Average water intake per animal (litres)}} \times \text{body weight (kg) of the animals to be treated} = \dots \text{ g of powder per litre of drinking water}$$

CONTRAINDICATIONS:

Do not use in cases of known hypersensitivity to doxycycline other tetracyclines or the excipient.

Do not use in case of known resistance to tetracyclines.

Do not use in animals with renal or hepatic disorders.

Do not use in ruminating animals.

WARNINGS / PRECAUTIONS:

The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of drinking water, calves and pigs should be treated parenterally.

Special precautions for use in animals

Due to the likely variability (time, geographical) in the occurrence of resistance of bacteria for doxycycline, bacteriological sampling and susceptibility testing of micro-organisms from diseased animals are highly recommended.

A high resistance rate for *Escherichia coli* isolated from chickens against tetracyclines has been documented. Therefore, the product should be used for the treatment of infections caused by *E. coli* only after susceptibility testing has been carried out.

Official and local antimicrobial policies should be taken into account when the product is used.

As the eradication of target pathogens may not be achieved, medication should therefore be combined with good animal management practices e.g. good hygiene, proper ventilation, no overstocking.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

People known to be allergic to the tetracycline class of antibiotics, should take special care when handling this product or the medicated solution.

This product may cause contact dermatitis and/or hypersensitivity reactions following skin or eye contact (with the powder or medicated solution), or if the powder is inhaled.

Take measures to avoid producing dust when incorporating the product into water.

During preparation and administration direct contact of the product with the skin, eyes and mucous membranes and inhalation of dust particles should be avoided.

Wear impervious gloves (e.g. rubber or latex), safety glasses and an appropriate dust mask (either a disposable half-mask respirator or a non-disposable respirator) when handling the product.

In the event of eye or skin contact, rinse the affected area with plenty of clean water.

If symptoms develop following exposure, such as skin rash, seek medical advice and show this warning to the physician. Swelling of the ace, lips or eyes or difficulty breathing are more serious symptoms and require urgent medical attention.

Do not smoke, eat or drink while handling the product.

Wash hands and contaminated skin immediately after handling the product.

DRUGS INTERACTIONS:

Divalent or trivalent cations (Mg, Fe, Al, Ca) can chelate tetracyclines. Tetracyclines should not be administered with antiacids, gels based on aluminum, preparations based on vitamins or minerals, since insoluble complexes are formed, which reduces the absorption of the antibiotic.

Do not use in conjunction with bactericidal antibiotics, such as penicillins or cephalosporins.

Doxycycline increases the action of anticoagulants.

PREGNANCY AND LACTATION:

Doxycycline showed no evidence of teratogenic or embryotoxic effects in laboratory animals.

In mammals, doxycycline crosses the placental barrier. Due to a lower affinity for calcium, doxycycline results in less staining of teeth compared to tetracycline. Doxycycline is found in breast milk.

The safety of the veterinary medicinal product has not been established during pregnancy and lactation. In pregnant and lactating animals use only accordingly to the benefit-risk assessment by the responsible veterinarian.

ADVERSE EFFECTS:

As for all tetracyclines, adverse reactions may occur, such as gastrointestinal disturbances and, with lower frequency, allergic reactions and photosensitisation.

If suspected adverse reactions occur, treatment should be discontinued.

OVERDOSE AND TREATMENT:

Not described. If suspected toxic reactions occur, the medication should be discontinued and appropriate symptomatic treatment should be initiated.

STORAGE CONDITIONS: Store at room temperature below 30°C. Protect from heat and sunlight exposure.

WITHDRAWAL PERIOD(S):

Meat and offal:

- Calves: 14 days

- Pigs: 6 days

- Chicken: 7 days

- Turkeys: 12 days

Eggs: Not authorized for use in laying birds producing eggs for human consumption.

Do not use within 4 weeks of onset of the laying period.

DOSAGE FORMS AND PACKAGING AVAILABLE: 500gm soluble powder packed into aluminium pouch.

DATE OF REVISION OF PACKAGE INSERT: 03 December 2018.



Product Registration Holder:
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